



Full-Stack A.I Engineer Technical Task: AI-Powered CV Screener

1. Introduction & Goal

Welcome! We've designed this technical task to be a practical and interesting challenge. The goal is to build a simple, end-to-end prototype of an AI-powered tool for screening CVs.

This task will allow you to demonstrate your ability to integrate AI models, handle data processing pipelines, and build a functional user interface. We are most interested in your ability to learn, problem-solve, and deliver an interesting working solution. Even if this is a new territory for you, we're excited to see your approach.

2. The Core Task

Your mission is to build a chat application that allows a user to ask questions about a collection of résumés (CVs). The project consists of three main parts:

1. Data Generation: Create a small dataset of 25-30 realistic-looking, fake CVs in PDF format.
2. Backend & AI Workflow: Develop a simple workflow to process these PDFs and make their content searchable and understandable by a Large Language Model (LLM). This is commonly known as a Retrieval-Augmented Generation (RAG) pipeline.
3. Frontend Chat Interface: Create a simple, functional web interface where a user can ask questions and get answers from the LLM based on the content of the CVs.

3. Requirements & Functionality

Core Requirements:

- **CV Generation:**
 - Generate 25-30 unique and fake CVs.
 - The CVs must be in PDF format.

- They should appear realistic, including elements like an AI-generated photo, contact information, work experience, skills, and education sections.
- The roles and languages used in the CVs are up to you.
- **Use any LLM to generate the texts and images.**
- **RAG Workflow:**
 - The system must extract text from the provided PDF documents.
 - It must process and store this information in a way that an LLM can retrieve it.
 - Optional: Grounded on the data only from the CVs.
 - **Use any suitable tool or solution for this.**
- **Chat Interface:**
 - A clean and simple user interface with a text input for questions and a display area for answers.
 - The LLM's responses must be based on the information contained within the CVs.
 - Optional: Source Indication: In the chat response, indicate which CVs were used as the source for the answer.
 - **Use any suitable tool or solution for this.**

4. Technical Guidelines & Suggestions

- **API Key:** You can choose any model you like suitable for the task. If necessary, you can start with free API keys from:
 - Google AI Studio: <https://aistudio.google.com/apikey> (free with limited use)
 - Openrouter: <https://openrouter.ai/settings/keys> (many models are free)
- **Creative Freedom:** You have complete freedom to choose your technology stack. The goal is a working product, not to follow a rigid set of rules. Don't feel the need to over-engineer the solution.
- **Hosting:** The final application does not need to be deployed or hosted online. Running it locally is perfectly fine.

5. Deliverables

Please provide the following:

- **A Short Video Demonstration (under 5 minutes):** Record your screen using [Loom](#) or a similar tool to walk us through your solution:
 - **The Process:** Briefly describe your CV generation pipeline and present examples of the resulting CVs and the Backend & AI Workflow.

- **The Demo:** Demonstrate the final application by asking sample questions (e.g., "Who has experience with Python?", "Which candidate graduated from UPC?", "Summarize the profile of Jane Doe.").
- **The Technical Highlight:** Conclude by showcasing the specific part of the code or logic you are most proud of. Tell us about the most challenging aspect you faced, a particularly creative solution you implemented, or an interesting approach you took. Help us value your technical depth by focusing on what makes your solution unique.
- Source Code: Include the source code of the project. GitHub repository is a great way for us to see your work.
- Optional: Overview diagram of this complete workflow.

6. Evaluation Criteria

We will be looking at your submission holistically, focusing on:

- **Execution & Functionality:** Does the application work as described?
- **Thought Process:** Your explanation of the architecture and technology choices.
- **Code Quality:** The clarity, structure, and readability of your code.
- **Creativity & Ingenuity:** Clever solution to a tricky problem, or implement a specific feature in an original way.
- **AI Literacy:** Your awareness of the relevant tools, models, and trends in the AI industry.
- **Learn & Adapt:** Your ability to tackle a new problem domain and produce a functional result is the most important factor.

7. Timeline

Please submit your deliverables within **2 days** from receiving this task.

If you have any questions, please don't hesitate to reach out to us. Good luck, and we look forward to seeing what you build!